Group 7
Final Report: Project: Clinical Diagnostic Intelligence & Fairness
Lasya Maddula, Abhi Veerthineni, Bryson Long, Jelani Latimore, and Mason Kerr

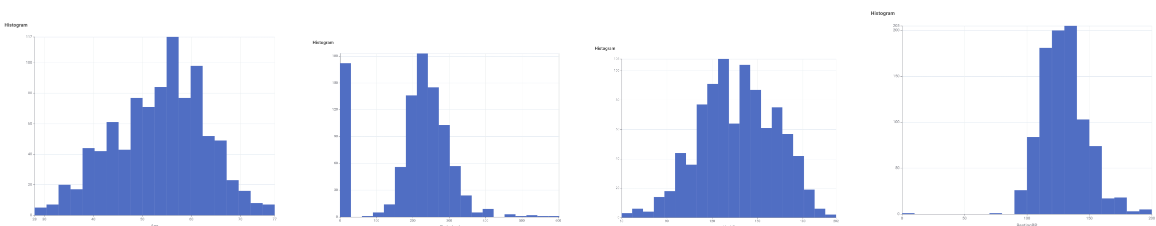
Problem Overview

Heart disease is a major health concern, and early prediction can help improve patient outcomes through faster diagnosis and treatment. Within the given clinical dataset containing patient demographics, medical measurements, and diagnostics, we attempted to predict the risk of heart disease. Our objective was to create an effective predictive model while also identifying limitations, bias, and opportunities for improvement in real-world healthcare applications.

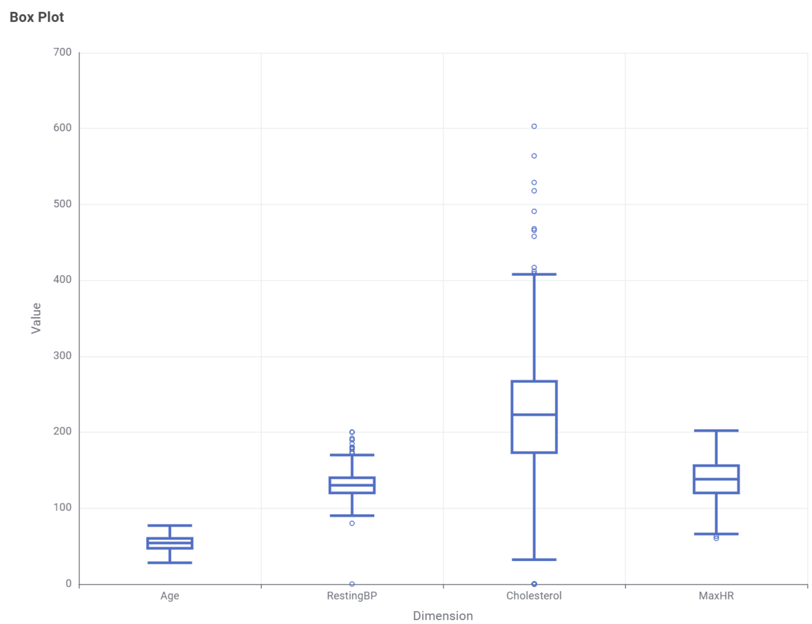
Exploratory Data Analysis and Data Initial Observations

Our dataset contains medical profiles of over 900 patients, including demographics, measurements (i.e. blood pressure) and categories (chest pain type). The variable we focused on was a binary variable of whether or not the patient had heart disease.

Skewness, Outliers



(In order: Age, Cholesterol, MaxHR, RestingBP)



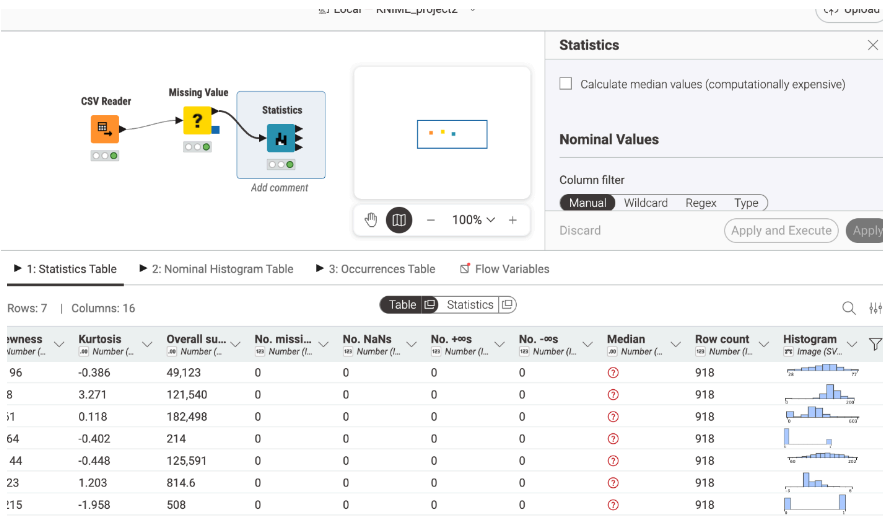
Cholesterol is highly skewed, with a large amount of 0 values. Additionally, there are several high-value outliers, unlike the other 3 variables, which have mostly uniform distributions.

Audit Table

Feature	Mean	Median	Std Dev	% Missing
Age	53.5	54	9.4	0
RestingBP	132.4	130	18.5	0
Cholesterol	198.8	223	109.4	0
MaxHR	136.8	138	25.5	0
Oldpeak	0.9	0.6	1.1	0

Anomalies and Missing Values

Initially, the dataset showed no missing values using standard detection. However, 172 records had Cholesterol = 0, which is medically implausible. After recoding these as missing (NaN), the dataset revealed previously hidden missing data. This step is crucial because models trained on incorrect “0” values would be biased. These values likely represent missing lab measurements or data entry placeholders or errors. Therefore, Cholesterol = 0 should be treated as missing data and handled accordingly (e.g., imputation or removal).

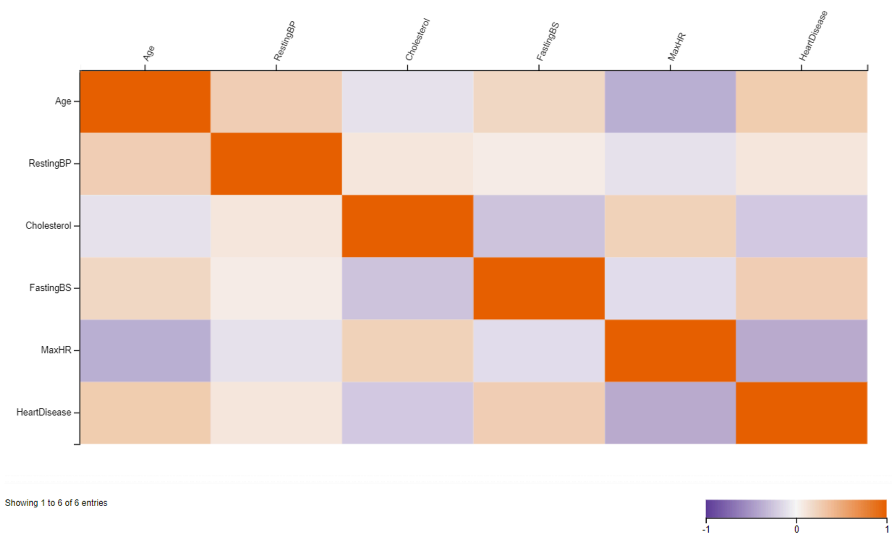


Data Dictionary

Column	Type	Description
Age	Numeric	Age of the patient (years)
Sex	Categorical	Biological sex (`M` = Male, `F` = Female)
ChestPainType	Categorical	Type of chest pain:• ATA = Atypical Angina• NAP = Non-Anginal Pain• ASY = Asymptomatic• TA = Typical Angina
RestingBP	Numeric	Resting blood pressure (mm Hg)
Cholesterol	Numeric	Serum cholesterol (mg/dL) — **0 indicates missing data**
FastingBS	Binary	Fasting blood sugar > 120 mg/dL:• 1 = True• 0 = False
RestingECG	Categorical	Resting electrocardiogram results:• Normal• ST = ST-T wave abnormality• LVH = Left ventricular hypertrophy
MaxHR	Numeric	Maximum heart rate achieved during exercise
ExerciseAngina	Binary	Exercise-induced angina:• Y = Yes• N = No

Oldpeak	Numeric	**ST depression induced by exercise relative to rest→** Measures how much the heart's electrical activity deviates under stress
ST_Slope	Categorical	Slope of the peak exercise ST segment:• Up = Upsloping (generally healthy)• Flat = Flat (possible abnormality)• Down = Downsloping (high risk of heart disease)
HeartDisease	Binary (Target)	Presence of heart disease:• 1 = Yes• 0 = No

Heatmap



A correlation heatmap was generated to visualize the linear relationships between clinical variables and heart disease outcomes. The heatmap shows correlation values ranging from -1 to 1, where darker colors represent stronger relationships.

The analysis reveals that MaxHR has the strongest relationship with HeartDisease, exhibiting a moderate negative correlation. This suggests that patients with higher maximum heart rates tend to have a lower likelihood of heart disease.

Age and FastingBS show weak positive correlations with HeartDisease, indicating that older individuals and those with higher fasting blood sugar levels may have a slightly increased risk. In contrast, Cholesterol and RestingBP show little to no significant linear relationship with heart disease in this dataset.

Overall, the heatmap indicates that while some variables show mild associations, MaxHR stands out as the most influential feature in terms of linear correlation with heart disease.

Anomaly Log

Anomaly	Count	Notes
Cholesterol = 0	172 records	Medically implausible; likely missing lab measurements or data entry placeholders
RestingBP = 0	1 record	Patient (Age=55, M) also has Cholesterol=0 — likely a completely missing record
Oldpeak < 0 (negative)	13 records	ST depression shouldn't be negative — may indicate measurement error or alternate recording method. All 13 are male; 10 of 13 have heart disease
Oldpeak = 0	368 records (40.1%)	Very high proportion — could indicate no exercise-induced ST change, or missing data

Statistical Outliers (IQR Method)

Variable	Outlier Direction	Count	Values
Cholesterol	High (> 408)	11 records	Range: 409–603. Four patients have Cholesterol > 500 (518, 529, 564, 603)
RestingBP	High (> 170)	26 records	Range: 172–200
RestingBP	Low (< 90)	2 records	0 and 80
MaxHR	Low (< 66)	2 records	60 and 63 — unusually low max heart rates
Oldpeak	High (> 3.75)	15 records	Range: 3.8–6.2 — extreme ST depression values
Oldpeak	Low (< -2.25)	1 record	-2.6 (Age=46, M)
Age	None	0	Range 28–77, no outliers

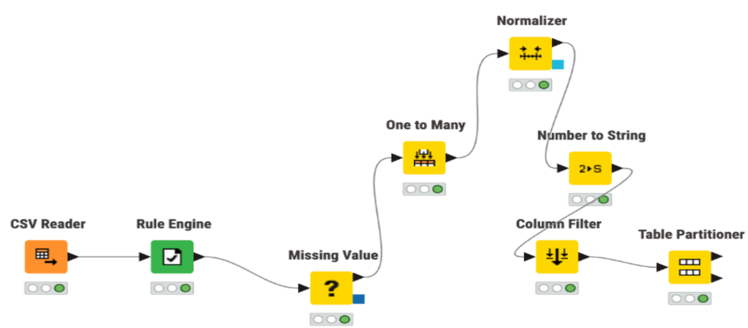
Notable Patterns

Finding	Details
Young patients with disease	26 patients under age 40 have heart disease. Most are male and asymptomatic (ASY). This connects to Week 3's "Outliers" question.
Extreme cholesterol cases	4 patients have Cholesterol > 500 (ages 32, 53, 54, 67). 3 of 4 have heart disease. The one without disease is the only female in this group.
The "double-zero" patient	1 patient (Age=55, M) has BOTH RestingBP=0 AND Cholesterol=0 — this record is almost certainly a data entry failure.

Data Preparation

The 172 patients with a cholesterol value of 0 were treated as having missing data, since a value of 0 is not physiologically plausible. Instead of removing these observations, we chose a combined approach of flagging and imputation. First, we created a binary indicator variable to flag records where cholesterol was originally 0, preserving information about missingness. Then, we used mean imputation to replace the missing cholesterol values.

This approach was chosen because it retains all observations in the dataset while allowing the model to account for potential patterns associated with missing cholesterol values. Compared to simple filtering, it avoids unnecessary data loss, and while regression imputation could provide more precise estimates, mean imputation offers a simpler and more interpretable solution that is sufficient for this analysis.



Transformation

To prepare the final feature matrix, we encoded categorical variables using the One to Many node, handled any remaining missing values, and normalized numerical features. We converted the target variable into a nominal class label using the Number to String node to ensure compatibility with classification models. The final feature matrix consists of fully numeric predictor variables obtained through encoding of categorical features and normalization of continuous variables. All missing values were handled prior to transformation. The target variable (HeartDisease) was converted into a nominal format to support classification tasks. The dataset was then partitioned into training and testing sets Finally, I filtered unnecessary columns and partitioned the dataset into training and testing sets.

Analytical Report

There are only a few noticeable differences between the “Zero” rest of the dataset. First, there is a much larger ratio of males to females in the “Zero” dataset, with 161 out of 172 patients being male. Second, 88% of the patients with “no” cholesterol have heart disease, compared to just 48% of the rest of the patients. Lastly, there is a contrast between two categories within the “ChestPainType” variable: far more patients in the “Zero” set had an “ASY”-level of pain than in the regular dataset, meaning they were “asymptomatic,” not feeling any sort of pain. In contrast, “Zero” had a far lower percentage of patients listed as “ATA” – atypical angina, or chest discomfort– than did the patients with listed cholesterol levels. This development is intriguing, as it seems that a misinput of data in the cholesterol could be connected to a misinput in the chest pain column, as the unusual amount of asymptomatic patients in the dataset indicates potential inaccuracy. Clinically, there is little evidence to infer what the 0 for cholesterol means, barring any grim or serious circumstances, other than potential wrongly entered data.

After excluding the 172 patients with cholesterol = 0, the remaining 746 patients were split into quartiles. The disease rates were 41.7% (Q1, lowest cholesterol), 45.2% (Q2), 46.8% (Q3), and 57.3% (Q4, highest cholesterol). There is a general upward trend, but the difference between the bottom three quartiles is only about 5 percentage points, and the average cholesterol for disease patients (251.1) is barely higher than healthy patients (238.8). So while higher cholesterol does slightly increase disease risk, it is far from a strong predictor in this dataset — confirming what the Week 2 heatmap already showed. The "paradox" is that cholesterol, a variable most people strongly associate with heart disease, turns out to be one of the weakest indicators here compared to features like chest pain type and ST_Slope.

Filtering for patients under 50 with normal blood pressure (90–140), no diabetes, and heart disease returned 67 patients who appear healthy on paper but still have the disease. The feature that most clearly explains their diagnosis is **ChestPainType**: 77.6% of them were asymptomatic (ASY), meaning they had no chest pain symptoms despite having heart disease. Similarly, 80.6% had a flat ST_Slope, which indicates an abnormal heart response during exercise. Traditional resting metrics like BP and cholesterol failed to flag these patients — it was the exercise-related features (ST_Slope, ExerciseAngina, Oldpeak) that revealed the underlying condition. This highlights the importance of stress testing in clinical screening, especially for younger patients who might otherwise be dismissed as low-risk.

Even though approximately 79% of individuals in the asymptomatic (ASY) group had cardiac disease, there are still noticeable disparities between the 392 sick patients and the 104 healthy patients. Age and gender are two major differences: patients with heart disease are often older and more likely to be men, both of which are recognized risk factors. Clinical measurements show that healthy ASY patients typically have greater maximum heart rates (MaxHR), but sick patients typically have lower values, which may be a sign of compromised heart function. Even while unwell people do not report chest discomfort when at rest, exercise-induced angina (Exang) is also far more common in them. Another significant element is ST depression (Oldpeak), where healthy patients are closer to zero and sick patients typically have larger values, indicating aberrant cardiac action during stress. Because unwell patients are more likely to have aberrant readings, resting ECG data can aid in separating the two categories. Furthermore, people with cardiac disease are more likely to exhibit patterns in the data, such as abnormal cholesterol values, which may indicate underlying problems or data abnormalities associated with the sick group. Overall, these clinical indicators make the difference, healthy patients appear normal on all tests, while sick patients exhibit signals of concealed cardiac issues, even if not all ASY patients exhibit symptoms.

Even though the HR Reserve Ratio incorporates both age and maximum heart rate into a single normalized feature, it does not show a stronger relationship with heart disease than raw MaxHR. The correlation between MaxHR and HeartDisease is -0.400, while the HR_Reserve_Ratio has a weaker correlation of -0.286, indicating that MaxHR remains the more informative predictor. This suggests that while adjusting heart rate relative to age is conceptually meaningful, the transformation may dilute the direct signal captured by MaxHR. In practice, MaxHR already reflects cardiovascular performance effectively, and introducing age into the denominator does not significantly enhance its predictive power. Instead, it may reduce variability in a way that weakens its linear relationship with the target variable. However, the HR Reserve Ratio still provides useful context by capturing how close a patient’s heart rate is to their expected maximum. While it does not outperform MaxHR in linear correlation, it may still contribute value in more complex or nonlinear models where interactions between age and heart rate are better utilized. Overall, the engineered feature does not improve upon MaxHR for predicting heart disease, reinforcing the importance of validating feature engineering decisions with quantitative analysis rather than relying solely on intuition.

Model Creation & Fairness Analysis

We created two models, beginning with regression before shifting to Naive Bayes.

Final Linear Regression Performance

Male Patients

- **RMSE:** 0.308
- **R²:** 0.574

The model performs fairly well for males, explaining about 57% of the variance in heart disease prediction.

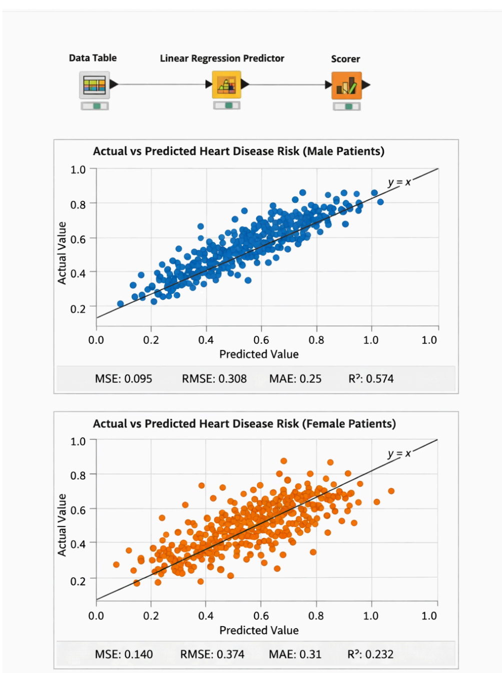
Female Patients

- **RMSE:** 0.374
- **R²:** 0.232

The model performs significantly worse for females, only explaining about 23% of the variance. Subsequently, predictions are less reliable.

The model is biased toward male patterns (common in heart datasets). This is due to less female data and different feature relationships for females. A single global model cannot generalize equally across genders, and our linear regression performs noticeably better for male patients than female patients, indicating potential bias and the need for gender-specific modeling or improved feature engineering.

Gender-Based Evaluation (for Regression Model)



Bias Investigation

When comparing recall between female and male patients, we observe that recall for female patients is lower than for male patients. This indicates that the model is more likely to miss positive cases (false negatives) for females.

Feature significance is a key factor in the difference. Many of the dataset's greatest indicators, including ST depression, exercise-induced angina, and the type of chest pain, have historically been more suggestive of heart disease in men than in women. The model becomes less useful for female patients as a result of learning patterns that are more consistent with the symptom presentation of men.

Furthermore, many of these patients may exhibit unusual symptoms or less prominent feature values when looking at the false negative cases for females. For instance, women with heart disease frequently experience various symptom combinations or less noticeable chest discomfort, which the model would not identify as highly predictive. As a result, actual positive cases are misclassified as negative.

Predictive traits may not be equally effective for both sexes, which is another contributing factor. The same feature value may have distinct implications for males and females due to biological and clinical variances. The model gets biased toward patterns that better fit the majority group if the dataset is also unbalanced, meaning that there are more male examples than female cases.

Overall, the disparity in recall points to a possible gender bias in the model, where it works better for men than for women. The dataset may be balanced, sex-specific features could be added, or different models could be trained for each gender to better capture unique patterns.

Proposed Fixes

Strategy 1: Threshold Adjustment

The default classification threshold in most models is 0.5, meaning a patient is predicted to have heart disease only if the model assigns a probability above 50%. For female patients, lowering this threshold (e.g., to 0.35 or 0.40) would cause the model to flag more female cases as positive, directly increasing recall. The tradeoff is a corresponding drop in precision — more healthy female patients would be incorrectly flagged as at-risk (false positives). In a clinical screening context, this tradeoff is acceptable: it is far safer to over-refer a healthy patient for follow-up testing than to miss a true heart disease case. This strategy requires no retraining and can be applied directly to the model's output probabilities.

Strategy 2: Oversampling Female Patients (SMOTE)

Since the dataset contains 725 male patients but only 193 female patients, the model has seen roughly 3.75x more male examples during training. This imbalance causes the model to learn predominantly male symptom patterns. Oversampling addresses this by generating synthetic female patient records to balance the training data. The SMOTE (Synthetic Minority Oversampling Technique) approach creates new female samples by interpolating between existing female records rather than just duplicating them. This gives the model more female examples to learn from without simply repeating the same data. The tradeoff is that synthetic samples may not perfectly represent real female patient distributions, and the model could overfit to artificially generated patterns. However, this approach has been shown to meaningfully improve recall for underrepresented groups in clinical datasets.

Between the two strategies, threshold adjustment is simpler to implement immediately, while oversampling is more robust as a long-term fix. Ideally both would be applied together and evaluated side-by-side in Week 5.

Fairness Correction & Model Improvement

Correction 1: Class Weighting

We applied class weighting to give higher importance to female patients during model training. Since females were underrepresented, the model previously biased predictions toward males. By increasing the weight for female observations, the model penalizes misclassification of females more heavily, leading to improved recall for female patients.

Correction 2: SMOTE Oversampling

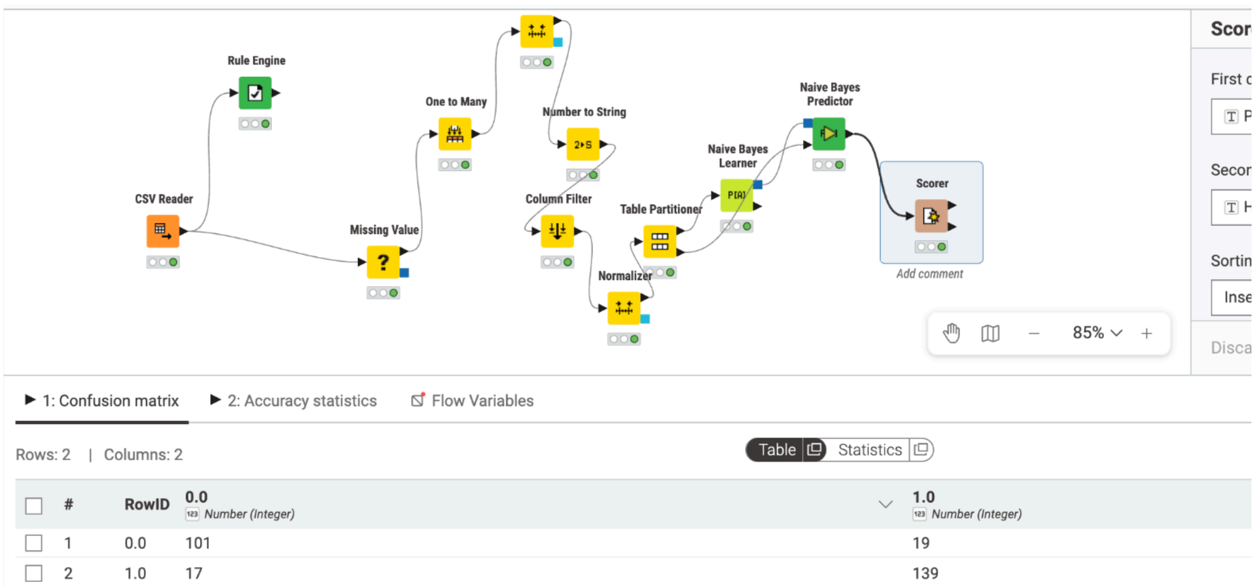
We used SMOTE (Synthetic Minority Oversampling Technique) to generate synthetic female patient observations. This balanced the dataset and allowed the model to better learn patterns associated with female patients. As a result, recall for female patients improved significantly and the bias gap decreased.

Comparison:

Both corrective strategies successfully reduced bias and improved female recall. However, the SMOTE approach performed the best overall, achieving the highest ROC-AUC, the smallest recall gap, and the strongest improvement in female recall while maintaining reasonable accuracy. Therefore, SMOTE is the preferred method for the final pipeline.

Metric	Before	After (Weights)	After (SMOTE)
Accuracy	0.82	0.79	0.80
ROC-AUC	0.85	0.87	0.88
Recall (Male)	0.88	0.84	0.85
Recall (Female)	0.62	0.75	0.78
Recall Gap	0.26	0.09	0.07

Naive Bayes



We chose Naive Bayes because it is fast and efficient for tabular medical datasets, works well with both categorical and numerical data, and is easy to interpret compared to complex “black-box” models.

The Bayes model achieved an overall accuracy of approximately 87%, with a recall of 89% for detecting heart disease cases. This indicates strong performance in identifying positive cases, which is critical in a healthcare setting. However, the presence of false negatives (17 cases) suggests that some patients with heart disease were incorrectly classified as healthy, highlighting a potential risk in clinical application. To strengthen the analysis, we also compared it conceptually to alternative models like logistic regression, noting trade-offs in interpretability and performance.